Joseph R. Farah

Miller Fellow of Astrophysics

Contact Information

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Citations: Google Scholar profile

Research Interests

I lead programs to measure the spins of compact objects across regimes, from infant millisecond magnetars in superluminous supernovae (SNe) to supermassive black holes resolved at the horizon scale. These measurements probe strong-field physics where no laboratory can reach and expose the engines driving the most energetic phenomena in the Universe.

Tags:

Astrophysical Phenomena: core-collapse SNe, superluminous SNe, black hole spacetimes, chirping SNe, photon rings
Observational Techniques: very-long-baseline interferometry (VLBI), radio astronomy, optical photometry, optical spectroscopy
Computational & Theoretical Methods: general relativity, Bayesian modeling, machine/deep learning, geometric modeling, interferometric imaging, shock-cooling models, light curve modeling

Education

University of California Santa Barbara, Santa Barbara, CA, USA 2021–2026
Ph.D. in Physics (completed) 2026
M.A. in Physics (completed) 2023

University of Massachusetts Boston, Boston, MA, USA 2017–2021
B.S. in Physics, summa cum laude, with distinction

Publication Summary

10	159	28,000+	45	105
First-author	*Total*	*Citations*	*h-index*	*i10-index*

Selected Publications

J. R. Farah, Prust, L. J., Howell, D. A., *et al.*. 2026. *Lense-Thirring precessing magnetar engine drives a superluminous supernova.* Nature, 651, 321-325.
Reported the discovery of the first "chirping" supernova and its implications for the central engine hypothesis. Selected for the cover of the March 12, 2026 issue of *Nature*.

J. R. Farah, Galison, P., Akiyama, K., *et al.*. 2022. *Selective Dynamical Imaging of Interferometric Data.* ApJL, 930, L18.
Seventh paper of the EHT's first Sagittarius A* results series, with full Collaboration co-authorship. Method recognized by the LeRoy Apker Award, EHT Early Career Award.

Selected Honors and Awards

Miller Fellowship, Miller Institute for Basic Research in Science, Berkeley 2026–2029
Award amount: $354,000

Bhaumik Prize Postdoctoral Fellowship, UCLA (declined) 2026
Award amount: $276,000

UC Santa Barbara Mananya Tantiwiwat Award 2025

Broida-Hirschfelder Fellowship 2025
CITATION: *In recognition of his project uncovering the secrets of the strangest object in the universe, the black hole photon ring.*

1st Place Award, UC Santa Barbara Grad Slam 2025
Round 1 (Physics and Chemistry)

UC Santa Barbara GSA Excellence in Teaching Award (nominated) 2025

UC Santa Barbara Academic Senate Outstanding Teaching Assistant Award (nominated) 2024

Event Horizon Telescope Early Career Award 2022
CITATION: *For his leadership and contribution to the dynamic imaging of the interferometric data used to image Sgr A*.*

LeRoy Apker Award 2021
CITATION: *For the invention of the selective dynamical imaging method, with applications for studying rapidly-varying black holes.*

UC Santa Barbara James and Mary Jo Hartle Graduate Fellowship 2021

National Science Foundation Graduate Research Fellowship 2021–2026
AWARD AMOUNT: $138,000

Breakthrough Prize in Fundamental Physics (co-recipient) 2020

Barry M. Goldwater Scholarship 2019

National Science Foundation Diamond Achievement Award‡ 2019

Smithsonian Fellowship 2018, 2019

Oracle Fellowship 2018, 2019

AJAS/AAAS Lifetime Fellowship 2018

Selected undergraduate scholarships: Chancellor's Scholarship (2017–2021), Alton J. Brann Endowed Scholarship (2019), Northrop Grumman Scholarship (2020), Arthur W. Martin III Scholarship (2021) 2017–2021

Selected Work and Research Experience

Miller Institute for Basic Research in Science
Miller Research Fellow, UC Berkeley — Berkeley, CA 2026–Present
Hosts: Prof. Daniel Kasen, Prof. Raffaella Margutti

Black Hole Initiative
Black Hole Explorer Fellow, Harvard University — Cambridge, MA 2026
Supervisor: Dr. Michael Johnson

Las Cumbres Observatory 2021–2026
NSF Graduate Research Fellowship — Goleta, CA
Supervisor: Prof. Andy Howell

Harvard-Smithsonian Center for Astrophysics 2018–2021
Smithsonian Fellowship with the Event Horizon Telescope — Cambridge, MA
Supervisor: Dr. Michael Johnson

Quantum Computing Research Group 2019–2021

Research Intern with Dr. Alioscia Hamma — Boston, MA

Harvard Laboratory for Particle Physics and Cosmology 2017–2018
Research Intern with Dr. Melissa Franklin — Cambridge, MA

QBism Research Group 2017–2019
Oracle Fellow with Dr. Christopher Fuchs — Boston, MA

Leadership and Collaboration Roles

Black Hole Explorer (BHEX)
Deputy Science Lead, Goal I (Photon Rings) 2026–Present
Core Member 2020–Present
Contributing to the science case of the proposed NASA space-VLBI mission since 2020. Lead development of its black hole spin-inference capabilities (i.e., photon ring analyses, cross-validation, and simulated data challenges) with extended line of first-author BHEX papers. Advise the PI on development of the BHEX Student Collaboration program.

CHIRPS Collaboration 2026–Present
Founder and Coordinator
CHaracterization and Inference of Rotating, magnetar-Powered SLSNe. Multi-institution collaboration (17 members across 9 institutions) for the discovery and modeling of chirping superluminous supernovae. Operates a public real-time chirp-detection filter on the ANTARES broker for Rubin/LSST alerts and leads the development of theoretical interpretations. Website.

Research Group, UC Berkeley 2026–Present
Group Lead
Supervise 5 student researchers at UC Berkeley, UC Santa Barbara, and Northwestern/CIERA on BHEX spin inference and the CHIRPS program.

UltraViolet EXplorer (UVEX) 2026–Present
Collaboration Member, Supernova Science
Assisting with development of the supernova observing strategy, including the cadence required to constrain early temperature evolution.

Global Supernova Project 2021–Present
Builder
Contributed to core collaboration infrastructure (including the Supernova Exchange) and performed data reduction for collaboration papers: 1000+ optical spectra and dozens of multi-band light curves. Can trigger Las Cumbres 1-m and 2-m photometric and spectroscopic follow-up.

Event Horizon Telescope Collaboration 2018–Present
Collaboration Member
Joined as the collaboration's first undergraduate member. Contributed to imaging, modeling, and theory for the first images of M87* (2019) and Sgr A* (2022). Led the selective dynamical imaging paper in the 2022 Sgr A* results (ApJL, 930, L18), recognized with the EHT Early Career Award.

Teaching and Mentoring

Invited workshop creator and lecturer, *Presentation presentation: a seminar on seminars*, Las Cumbres Observatory, Goleta, CA. January 2026.

Death Star Seminar, founder and lecturer. Informal graduate seminar series on supernova theory, Las Cumbres Observatory, Goleta, CA. Designed curriculum and delivered lectures. 2026.

Creator and lecturer, online derivation series, *Astraveo*. Blackboard lectures working through core astrophysics and general science results from first principles. 2025–Present.

Teaching Assistant, PHYS 23 (Electrostatics and waves), UC Santa Barbara. Fall 2024.

UC Santa Barbara GSA Excellence in Teaching Award (nominated)
UC Santa Barbara Academic Senate Outstanding Teaching Assistant Award (nominated)

Guest Lecturer. Astronomy 121, University of Massachusetts Boston, Boston, MA. 2019 (x2), 2020 (x2).

Graduate research mentoring

Nicole Flors (CIERA/Northwestern) 2026–Present

Project: Simulation of magnetar+disk system with HAM-R

Undergraduate research mentoring

Eyouel Abate (UC Berkeley) 2026–Present

Project: Development of automated CHIRPS filter on Rubin infrastructure

Austin Mei (UC Berkeley) 2026–Present

Frequency-domain methods for identification of periodicity in supernova light curves

Mai Haller (UC Berkeley) 2026–Present

Investigation of contrast ratios of SLSNe to surface brightness of hosts

Courtney Duong (UC Santa Barbara) 2024–Present
Project: Convolutional neural network for extraction of the $n = 1$ black hole photon ring (Published in *ApJ*)
Award: *Gene and Susan Lucas Undergraduate Research Scholarship*
Award: *Edison STEM Research Scholarship*

Franklin Myhre (UC Santa Barbara) 2024–Present
Project: Development of a `python` library for automated image-domain subring extraction

Sanjit Masanam (UC Santa Barbara) 2024–2026
Award: *2025 Goldwater Scholarship (nominated)*
Award: *2025 Ernest F. Hollings Scholarship*

Songgun Lee (UC Santa Barbara) 2024–2026
Co-founder at Masterminding

Tazzy Imbabi (UC Santa Barbara) 2024
Project: Exploration of selective dynamical imaging applications to BHEX

Service

Referee:
The Astrophysical Journal
Astronomy & Astrophysics

Internal referee, Event Horizon Telescope Collaboration.

Invited presenter, Las Cumbres Observatory Science Advisory Committee, Goleta, CA. Invited to present on the observatory's high-energy astrophysics science program. August 2025.

Local Organizing Committee, EHT Collaboration Meeting. 2020.

Local Organizing Committee, 2nd EHT Imaging Workshop, Black Hole Initiative, Harvard University, Cambridge, MA. July 2018.

Founder and organizer, *Research@UMB*, University of Massachusetts Boston. Connected undergraduates seeking research experience with faculty through faculty lightning-talk sessions and direct student–faculty matching. 2019.

Invited Talks and Colloquia

A new theoretical framework for chirping supernovae. Montana State University, Bozeman, MT. October 2026.

Precessing magnetar central engine drives a superluminous supernova. Galactic Center Group, UC Los Angeles, Los Angeles, CA. April 2026.

The first 4 years of SN 1993J revisited. UC Berkeley, Berkeley, CA. September 2025.

Lense-Thirring precessing magnetar engine drives a superluminous supernova. Multi-RAPTOR colloquium, UC Berkeley, Berkeley, CA. September 2025.

Inferring black hole spin from images of the photon ring. VandyGRAF Initiative, Vanderbilt University, Nashville, TN. September 2024.

What Lies Within: Imaging the Galactic Center. April APS Meeting, New York, NY. Invited session speaker. April 2022.

The First Image of a Black Hole. Skyscrapers, Inc., Seagrave Memorial Observatory, North Scituate, RI. 2021.

Seminars and Contributed Talks

A new theoretical framework for chirping supernovae. Multi-RAPTOR seminar, UC Berkeley, Berkeley, CA. October 2026.

Spin inference of compact objects across regimes. Black Hole Initiative, Cambridge, MA. August 2026.

Precessing magnetar central engine drives a superluminous supernova. METAL group, University of Virginia, Charlottesville, VA. August 2026.

Spin inference of compact objects across regimes. Greene research group, Princeton University, Princeton, NJ. July 2026.

Precessing magnetar central engine drives a superluminous supernova. Narayan group, Center for Astrophysics | Harvard & Smithsonian, Cambridge, MA. July 2026.

General relativity meets a supernova for the first time. Las Cumbres Observatory colloquium, Goleta, CA. January 2026. Event URL.

Lense-Thirring precessing magnetar engine drives a superluminous supernova. KIPAC, Stanford University, Stanford, CA. November 2025.

Machine- and deep-learning-driven angular momentum inference from Black Hole Explorer observations of the $n = 1$ photon subring. *Topics in Gravitational Physics*, APS Global Physics Summit. March 2025. Event URL.

Current and future impacts of survey capabilities on the stripped-envelope problem. Kavli Institute for Theoretical Physics, Santa Barbara, CA. February 2025.

Shock-cooling constraints via early-time observations of the Type IIb SN 2022hnt. Kavli Institute for Theoretical Physics, Santa Barbara, CA. February 2025.

Selective dynamical imaging of interferometric data and the second image of a black hole. *Astro Tea*, University of California, Santa Barbara, CA. June 2022.

Seeing the unseeable: the first image of a black hole. University of Massachusetts Boston, Boston, MA. May 2019.

Imaging a black hole with the Event Horizon Telescope. Harvard University, Cambridge, MA. April 2019.

Outreach and Public Engagement

Co-founder and host, *Astraveo*, science-communication YouTube channel (with D. James). 2023–

Present. Channel URL.
14,000+ subscribers; most recent video (including an on-camera blackboard lecture) 150,000+ views; 4+ videos with 25,000+ views each, 100+ videos total; research vlog, documentaries.

General relativity beats the heart of a dying star. Public lecture, Santa Barbara Museum of Natural History. March 2026. Event URL.

Extragalactic science with the Vera C. Rubin Observatory. Public lecture, Santa Barbara Museum of Natural History. August 2025. Event URL.

The Black Hole Explorer and the edge of the universe. Public lecture, Santa Barbara Museum of Natural History. April 2025. Event URL.

The Black Hole Explorer. *Astronomy on Tap*, Santa Barbara, CA. December 2024. Event URL.

What does an astronomer do? Canalino Elementary School, Carpinteria, CA. October 2024.

The biggest telescopes! Aliso Elementary School, Carpinteria, CA. October 2023.

Imaging the black hole at our galaxy's center. Public lecture, Santa Barbara Museum of Natural History. May 2023. Event URL.

LCO's top 9 tips for taking your best black hole photo. *Astronomy on Tap*, Santa Barbara, CA. November 2022. Event URL.

AskScience AMA Series: Event Horizon Telescope results on the Galactic Center. *Reddit*. May 2022. Article URL.

LCO's top 9 tips for taking your best black hole photo. Honors and AP physics classes, Medford High School, Medford, MA. 2022.

Selected Media and Press

Appearances

Guest, *Nature Podcast*, on the chirping supernova SN 2024afav. March 2026. Listen.
Participant, Event Horizon Telescope press conference unveiling the first image of Sgr A*, Washington, D.C. May 2022.
Galison, P. L. (Director). 2021. *Black Holes: The Edge of All We Know*. Sandbox Films. View on Netflix.
Galison, P. L. (Director). 2019. *Portrait of a Shadow*. Sandbox Films.

Coverage

SN 2024afav (Nature, 2026): Reuters, *The Times*, *The Independent*, *Science*, *Scientific American*, *Nature* News & Views, *Nature* Video, *Science News*, *Ars Technica*, *Smithsonian*, *Popular Science*, *Sky & Telescope*, *BBC Sky at Night*, *Space.com*, *Live Science*, *ScienceAlert*, *IFLScience*; international coverage including a print feature in *Tähdet ja avaruus* (Finland), *Spektrum der Wissenschaft* (Germany), Agencia SINC (Spain), *Media INAF* (Italy), *Der Standard* (Austria), and *La Nación* (Argentina); YouTube features by Astrum and Anton Petrov. Press releases by UC Berkeley, UC Santa Barbara, and Las Cumbres Observatory.

Selective dynamical imaging of Sgr A (2022):* profiles in the *Santa Barbara Independent* and on KCLU radio; UC Santa Barbara/LCO press release.

Event Horizon Telescope, undergraduate work (2019–2021): profile in *Symmetry* magazine (2020); Center for Astrophysics | Harvard & Smithsonian and UMass Boston coverage of the Apker Award (2021).

Publications (first-author)

Stats (all papers):
Citations: 28,000+
Ref'd Papers: 159
h-index: 45
i10-index: 105

10. **J. R. Farah**, Prust, L. J., Yu, H. 2026. *A universal relation predicts the observed reversal of a*

supernova chirp. arXiv:2610.39240.

9. **J. R. Farah**, Palumbo, D. C. M., Johnson, M. D., Chang, D. O. 2026. *Spin inference with the Black Hole Explorer. I. Fisher information matrix forecast*. arXiv:2608.23672.

8. **J. R. Farah**, Lupsasca, A., Quataert, E., Johnson, M. D. 2026. *Interferometric inference of black hole spin from photon ring size and brightness*. ApJ, 1003, 63.

7. **J. R. Farah**, Prust, L. J., Howell, D. A., *et al.*. 2026. *Lense-Thirring precessing magnetar engine drives a superluminous supernova*. Nature, 651, 321-325. **NB:** selected for cover of the March 12, 2026 issue.

6. **J. R. Farah**, Howell, D. A., Hiramatsu, D., *et al.*. 2026. *When IIb Ceases To Be: Bridging the Gap Between IIb and Short-plateau Supernovae*. ApJ, 998, 321.

5. **J. R. Farah**, Prust, L. J., Terreran, G., *et al.*. 2025. *The First 4 Years of SN 1993J Revisited: Geometric m-ring Modeling of the Radio Shell with Closure Quantities Only*. Submitted to ApJ, positive referee report received. arXiv e-prints, arXiv:2509.20601.

4. **J. R. Farah**, Davelaar, J., Palumbo, D., Johnson, M., Delgado, J. 2025. *Machine and Deep Learning-driven Angular Momentum Inference from BHEX Observations of the $n = 1$ Photon Ring*. ApJ, 983, 185.

3. **J. R. Farah**, Howell, D. A., Terreran, G., *et al.*. 2025. *Shock-cooling Constraints via Early-time Observations of the Type IIb SN 2022hnt*. ApJ, 984, 60.

2. [S] **J. R. Farah**, Galison, P., Akiyama, K., *et al.*. 2022. *Selective Dynamical Imaging of Interferometric Data*. ApJL, 930, L18. arXiv:2409.08321.

1. **J. R. Farah**, Pesce, D. W., Johnson, M. D., Blackburn, L. 2020. *On the Approximation of the Black Hole Shadow with a Simple Polar Curve*. ApJ, 900, 77.

Publications

(Major or normal contribution, * indicates student-led)

(last updated: Oct 2026)

149. Shrestha, M., Kwok, L. A., Sand, D. J., ..., **J. R. Farah**, *et al.* 2026. *JWST Observations of SN 2024abup: First Detection of CO in a Broad-lined Type Ic Supernova and Constraints on r-process Nucleosynthesis*. ApJL, 1009, L53. arXiv:2606.28561.

148. Macrie, C. W., Larison, C., Sears, H., ..., **J. R. Farah**, *et al.* 2026. *JWST Nebular Spectroscopy of SN 2023qov: Circumstellar Dust Emission in a Normal Type Ia Supernova*. ApJ, 1009, 241. arXiv:2604.09777.

147. Barna, B., Singh, M., Kwok, L. A., ..., **J. R. Farah**, *et al.* 2026. *Tomography analysis of the intermediate-luminosity Type Iax SN 2024pxl*. arXiv e-prints, arXiv:2609.28712.

*146. Duong, C., Myhre, F., **J. R. Farah**. 2026. *Convolutional Neural Network for Extraction of $n = 1$ Photon Ring of Black Holes*. arXiv e-prints, arXiv:2609.25344.

145. Dukiya, N., Misra, K., Pastorello, A., ..., **J. R. Farah**, *et al.* 2026. *Early NIR echo and the time variable mass loss history of Type IIn SN 2024kgi*. arXiv e-prints, arXiv:2609.25233.

[S] This was the seventh paper of the 10-paper first Galactic Center results series from the Event Horizon Telescope, presenting the second image of a black hole and the first horizon-scale images of Sgr A*.

144. Song, C.-Y., Jiang, N., Wang, X., ..., **J. R. Farah**, *et al.* 2026. *SN 2025fhm: A central-engine powered Ic-BL supernova associated with X-ray transient EP250304a.* arXiv e-prints, arXiv:2609.08356.

143. Dubey, M., Misra, K., Dukiya, N., ..., **J. R. Farah**, *et al.* 2026. *SN 2022xus: bridging the gap between Type IIP and IIL supernovae.* MNRAS, 551, stag1427. arXiv:2607.25323.

142. Hebbar, D., Roy, R., Nicholl, M., ..., **J. R. Farah**, *et al.* 2026. *SN 2019cqc: A Hydrogen-rich Superluminous Supernova Showing Electron-scattering Signatures.* ApJ, 1008, 135. arXiv:2608.30309.

141. Jiang, S.-Q., Xu, D., Lei, W.-H., ..., **J. R. Farah**, *et al.* 2026. *EP251023a: A fast X-ray transient featuring a magnetar-powered optical internal plateau followed by a steep decay.* A&A, 713, L5. arXiv:2606.24329.

140. Müller, H., von Fellenberg, S. D., Zeng, A.-L., ..., **J. R. Farah**, *et al.* 2026. *Dynamics and geometry of the inner sub-parsec-scale jet in 3C 279 observed with the Event Horizon Telescope.* A&A, 713, A274. arXiv:2607.16104.

139. Principe, G., Algaba, J. C., Aviano, E., ..., **J. R. Farah**, *et al.* 2026. *Broadband multiwavelength properties of the archetypal blazar 3C 279 during the 2017 Event Horizon Telescope campaign.* A&A, 712, A224. arXiv:2606.25868.

138. Aghakhanloo, M., Torres-Albà, N., Smith, N., ..., **J. R. Farah**, *et al.* 2026. *AT 2016blu: Accretion-Powered Outbursts in a Luminous Blue Variable and Compact Object Binary.* arXiv e-prints, arXiv:2608.20469.

137. Kwok, L. A., Blondin, S., Miller, A. A., ..., **J. R. Farah**, *et al.* 2026. *JWST Spectroscopy of Type Ia Supernova 2025rbs from Maximum Light to the Nebular Phase.* arXiv e-prints, arXiv:2608.10451.

136. Dong, Y., Villar, V. A., Nugent, A., ..., **J. R. Farah**, *et al.* 2026. *Spectral Diversity in Type Ibn Supernovae and the Large Host Offset of SN 2024acyl.* ApJ, 1007, 146. arXiv:2511.03926.

135. Srinivasaragavan, G. P., Li, D., Hall, X. J., ..., **J. R. Farah**, *et al.* 2026. *EP250827b/SN 2025wkm: An X-Ray Flash-supernova Powered by a Central Engine and Circumstellar Interaction.* ApJL, 1006, L36. arXiv:2512.10239.

134. Ransome, C. L., Sand, D. J., Bostroem, K. A., ..., **J. R. Farah**, *et al.* 2026. *The Eye of Sauron in SN 2025ngs: A Short-plateau Cousin of SN 1998S with Evidence for a Ring-like Circumstellar Medium.* ApJ, 1006, 230. arXiv:2605.05380.

133. Vasylyev, S. S., Hosseinzadeh, G., Dessart, L., ..., **J. R. Farah**, *et al.* 2026. *Ultraviolet to Infrared Spectroscopy of the Type Ibn SN 2023tsz Suggests a Lower-mass Progenitor.* arXiv e-prints, arXiv:2607.15342.

132. Hsu, B., Smith, N., Bostroem, K. A., ..., **J. R. Farah**, *et al.* 2026. *Evidence for Asymmetric Ejecta and Circumstellar Material in SN 2023ixf Inferred from Extensive Nebular-phase Observations.* arXiv e-prints, arXiv:2607.02859.

131. Hall, S., Kwok, L. A., Ravi, A. P., ..., **J. R. Farah**, *et al.* 2026. *JWST Observations of Calcium-Strong Transients: I. Complex Nebular He Emission in SN 2024uj.* arXiv e-prints, arXiv:2607.00111.

130. Rosowsky, M., Valenti, S., Ravi, A. P., ..., **J. R. Farah**, *et al.* 2026. *SN 2023rve: A Type II Supernova with No Nebular Oxygen.* arXiv e-prints, arXiv:2606.30979.

129. Ransome, C. L., Subrayan, B. M., Sand, D. J., ..., **J. R. Farah**, *et al.* 2026. *First Results from the LSST Shadow Survey: The Restless Luminous Blue Variable AT2017des in the Virgo-Cluster Galaxy, NGC4532.* arXiv e-prints, arXiv:2606.23784.

128. Keinan, I., Arcavi, I., Howell, D. A., ..., **J. R. Farah**, *et al.* 2026. *Las Cumbres Observatory Gravitational-wave Follow-up in the Third and Fourth Observing Runs: Strengths and Weaknesses of a Rapid Response Galaxy Targeted Strategy.* ApJ, 1004, 230. arXiv:2604.15129.

127. Cai, J., Zeng, X., Zhou, W., ..., **J. R. Farah**, *et al.* 2026. *SN 2019dks: A 99aa-like Type Ia Supernova with a Low-velocity Gradient.* ApJ, 1004, 98.

126. Bostroem, K. A., Valenti, S., Sand, D. J., ..., **J. R. Farah**, *et al.* 2026. *Late-time Hubble Space Telescope Ultraviolet Spectra of SN 2023ixf and SN 2024ggi Show Ongoing Interaction with Circumstellar Material.* ApJ, 1004, 23. arXiv:2508.11756.

125. Barna, B., Bánhidi, D., Szalai, T., ..., **J. R. Farah**, *et al.* 2026. *The extremely low-luminosity Type Iax SNe 2022ywf and 2023zgx.* A&A, 710, A72. arXiv:2602.23175.

124. Hiramatsu, D., Berger, E., Tsuna, D., ..., **J. R. Farah**, *et al.* 2026. *The pair-instability origin of supernova 2023vbw.* arXiv e-prints, arXiv:2605.16487.

123. Ravi, A. P., Kumar, S., Baer-Way, R., ..., **J. R. Farah**, *et al.* 2026. *The Double-peaked Calcium-strong SN 2025coe: Progenitor Constraints from Early Interaction and Ejecta Asymmetries.* ApJ, 1003, 33. arXiv:2601.00415.

122. Gerard, M., Andrews, J. E., Clayton, G. C., ..., **J. R. Farah**, *et al.* 2026. *SN 2024abfl: A Flat-plateau, Low-luminosity Type IIP Supernova with Early Circumstellar Matter Interaction.* ApJ, 1002, 70. arXiv:2601.02638.

121. Chen, L., Wang, X., Wu, Q., ..., **J. R. Farah**, *et al.* 2026. *SN 2024iss: A double-peaked Type IIb supernova with evidence of circumstellar interaction.* A&A, 710, A33. arXiv:2510.22997.

120. Könyves-Tóth, R., Bodola, Z., Szalai, T., ..., **J. R. Farah**, *et al.* 2026. *Distance estimate to NGC 6951 from supernova siblings Type IIP SN 2020dpw and Type Ib SN 2021sjt.* A&A, 709, A68. arXiv:2603.24769.

119. Aster, C., Inserra, C., Pastorello, A., ..., **J. R. Farah**, *et al.* 2026. *SN 2024abvb: a Type Ibn/Icn supernova with evidence of helium and an extreme offset from its host galaxy.* MNRAS, 547, stag404. arXiv:2602.20775.

118. Bernshteyn, V., Conroy, N. S., Bauböck, M., ..., **J. R. Farah**, *et al.* 2026. *Ring Asymmetry and Spin in M87*.* ApJ, 1000, 231. arXiv:2601.00394.

117. Georgiev, B., Tiede, P., von Fellenberg, S. D., ..., **J. R. Farah**, *et al.* 2026. *Locating the missing large-scale emission in the jet of M87* with short EHT baselines.* A&A, 709, A26. arXiv:2601.13356.

116. Albentosa-Ruiz, E., Washington, J. E., Marchili, N., ..., **J. R. Farah**, *et al.* 2026. *Full-polarization millimeter wavelength variability of Sagittarius A* during the 2018 EHT campaign.* A&A, 708, A179. arXiv:2604.10287.

115. Singh, M., Kwok, L. A., Jha, S. W., ..., **J. R. Farah**, *et al.* 2026. *Photometry and Spectroscopy of SN 2024pxl: A Luminosity Link among Type Iax Supernovae.* ApJ, 999, 227. arXiv:2505.02943.

114. Dubey, M., Misra, K., Csörnyei, G., ..., **J. R. Farah**, *et al.* 2026. *SN 2023zcu: A Type IIP*

SN with Early Flash Features. ApJ, 999, 93. arXiv:2601.14830.

113. Zimmerman, E. A., Gal-Yam, A., Groot, P. J., ..., **J. R. Farah**, *et al.* 2026. *A Faint Progenitor System for the Faint Supernova 2024vjm*. arXiv e-prints, arXiv:2602.09096.

112. Kumar, H., Blanchard, P. K., Berger, E., ..., **J. R. Farah**, *et al.* 2026. *SN 2024afav: A Superluminous Supernova with Multiple Light-curve Bumps and Spectroscopic Signatures of Circumstellar Interaction*. ApJL, 998, L3. arXiv:2512.06067.

111. Saurabh, Müller, H., von Fellenberg, S. D., ..., **J. R. Farah**, *et al.* 2026. *Probing jet base emission of M87* with the 2021 Event Horizon Telescope observations*. A&A, 706, A27. arXiv:2512.08970.

110. Gómez, J. L., Cho, I., Traianou, E., ..., **J. R. Farah**, *et al.* 2026. *Spatially resolved polarization swings in the supermassive binary black hole candidate OJ 287 with first Event Horizon Telescope observations*. A&A, 705, A23.

109. Event Horizon Telescope Collaboration, ..., **J. R. Farah**, *et al.* 2025. *Horizon-scale variability of M87* from 2017–2021 EHT observations*. A&A, 704, A91. arXiv:2509.24593.

108. Park, S. H., Rho, J., Yoon, S.-C., ..., **J. R. Farah**, *et al.* 2025. *Near-infrared spectroscopy and detection of carbon monoxide in the Type II supernova SN 2023ixf*. A&A, 703, A227. arXiv:2507.11877.

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(last updated: Aug 2025)

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Classifications, telegrams, and circulars

(last updated: Aug 2025)

First or co-author on 90+ Transient Name Server classification reports, AstroNotes, and GCN circulars, 2022–present, as part of the Global Supernova Project.

Software Development

`eht-imaging`. Contributor. Repository.

The Supernova Exchange (SNEx). Contributor. Live site link.